BSIT 375: Knowledge Discovery and Data Science

Week 3 Lecture Notes: Basic Probablity and Bayes Theorem

Based on Larose & Larose: *Discovering Knowledge in Data (2nd Ed)*

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Probability Foundations: From Basics to Bayes' Theorem

Introduction to Probability

Probability is the mathematical language for quantifying uncertainty. It is the foundation of statistics and machine learning. We begin with basic definitions.

Key Terms

- **Experiment**: A process that leads to an outcome. (e.g., rolling a die, flipping a coin, drawing a card)
- **Sample Space (S)**: The set of all possible outcomes of an experiment.
- **Event (E)**: A subset of the sample space, i.e., a collection of outcomes.
- **Probability ($P(E)$)**: A number between 0 and 1 that measures the likelihood that event E occurs.

Axioms of Probability: For any event E:

1. $0 \leq P(E) \leq 1$
2. $P(S) = 1$ (the sample space always occurs)
3. If $E_1, E_2, \ldots$ are mutually exclusive events (they cannot happen at the same time), then

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i).$$

Classical (Theoretical) Probability: If all outcomes in S are equally likely, then

$$P(E) = \frac{|E|}{|S|} = \frac{\text{Number of outcomes in } E}{\text{Total number of outcomes in } S}.$$

Example 1: Rolling a Die

Experiment: Roll a fair six-sided die.

- Sample space: $S = \{1, 2, 3, 4, 5, 6\}$, $|S| = 6$.
- Event A: rolling an even number: $A = \{2, 4, 6\}$, $|A| = 3$.
- $P(A) = \frac{3}{6} = 0.5$.

Counting Techniques

When sample spaces are large, we need efficient ways to count outcomes. The following techniques are essential.

Multiplication Principle (Product Rule): If an operation consists of k steps, where the first step can be done in n_1 ways, the second in n_2 ways, ..., and the k-th in n_k ways, then the entire operation can be done in $n_1 \times n_2 \times \cdots \times n_k$ ways.

Example 2: Menu Choices

A restaurant offers 3 appetizers, 5 main courses, and 4 desserts. How many different meals (one appetizer, one main, one dessert) can a customer order?

$$3 \times 5 \times 4 = 60 \text{ meals.}$$

Permutations: A permutation is an ordered arrangement of distinct objects. The number of permutations of n distinct objects taken r at a time is

$$P(n, r) = \frac{n!}{(n-r)!}.$$

Key phrase: “order matters.”

Example 3: Permutations

How many ways can 5 people be seated in a row of 5 chairs?

$$P(5, 5) = 5! = 120.$$

If only 3 chairs are available, the number of ways to seat 5 people in 3 chairs is

$$P(5, 3) = \frac{5!}{(5-3)!} = 5 \times 4 \times 3 = 60.$$

Combinations: A combination is an unordered selection of distinct objects. The number of combinations of n distinct objects taken r at a time is

$$C(n,r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

Key phrase: "order does not matter."

Example 4: Combinations

A committee of 3 is to be chosen from 10 people. How many different committees are possible?

$$\binom{10}{3} = \frac{10!}{3!7!} = 120.$$

Trick: When to Use Permutation vs Combination:

- If the problem mentions **arrangements**, **ordered**, **rankings**, **positions** $\rightarrow$ use permutation.
- If the problem mentions **selection**, **group**, **committee**, **subset** $\rightarrow$ use combination.

Probability Using Counting Techniques

Often, we compute probabilities by counting favorable outcomes and total outcomes.

Example 5: 3-Card Hand Probabilities (Teen Patti Style)

In a common 3-card game, we draw 3 cards from a standard 52-card deck without replacement. We want to find the probability of two specific "power hands":

1. **Trial (Set):** All three cards are of the same rank (e.g., three Kings).
2. **Color (Flush):** All three cards belong to the same suit (e.g., all Hearts).

1. Total Sample Space: The total number of ways to pick any 3 cards from 52:

$$n(S) = \binom{52}{3} = \frac{52 \times 51 \times 50}{3 \times 2 \times 1} = 22,100$$

2. Probability of a Trial: To get a Trial, we must choose one rank out of 13, and then choose 3 suits out of the 4 available for that rank.

$$n(\text{Trial}) = \binom{13}{1} \times \binom{4}{3} = 13 \times 4 = 52$$

$$P(\text{Trial}) = \frac{52}{22,100} \approx 0.00235 \text{ (or } 0.24\%)$$

3. Probability of a Color: To get a Color, we must choose one suit out of 4, and then choose 3 cards from the 13 available in that suit.

$$n(\text{Color}) = \binom{4}{1} \times \binom{13}{3} = 4 \times \frac{13 \times 12 \times 11}{3 \times 2 \times 1} = 4 \times 286 = 1,144$$

$$P(\text{Color}) = \frac{1,144}{22,100} \approx 0.05177 \text{ (or } 5.18\%)$$

Example 6: Lottery Probability

In a lottery, you pick 6 numbers from 1 to 49. What is the probability of winning the jackpot (matching all 6)?

$$\text{Total outcomes} = \binom{49}{6}, \quad \text{Favorable} = 1, \quad P = \frac{1}{\binom{49}{6}} \approx 7.15 \times 10^{-8}.$$

Independence, Dependence, and Sampling with/without Replacement

Two events A and B are **independent** if $P(A \cap B) = P(A)P(B)$. Otherwise, they are dependent.

Sampling Methods

- **With replacement**: After drawing an object, it is returned to the population before the next draw. Trials are independent.
- **Without replacement**: Objects are not returned. Trials are dependent.

Example 7: Balls from an Urn

An urn contains 3 red and 2 blue balls. Two balls are drawn.

1. With replacement:

$$P(\text{first red, second red}) = \frac{3}{5} \times \frac{3}{5} = \frac{9}{25}.$$

2. Without replacement:

$$P(\text{first red, second red}) = \frac{3}{5} \times \frac{2}{4} = \frac{6}{20} = \frac{3}{10}.$$

Conditional Probability

Conditional probability measures the probability of an event given that another event has occurred.

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0.$$

Multiplication Rule:

$$P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A).$$

Example 8: Conditional Probability

A family has two children. Given that at least one is a boy, what is the probability that both are boys?

- Sample space: $\{BB, BG, GB, GG\}$ (order matters).
- Event A: both boys $= \{BB\}$.
- Event B: at least one boy $= \{BB, BG, GB\}$.
- $P(A \mid B) = \frac{|A \cap B|}{|B|} = \frac{1}{3}$.

Bayes' Theorem

Bayes' Theorem provides a way to reverse conditional probabilities. It is fundamental in machine learning (e.g., Naïve Bayes classifier).

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}.$$

More generally, for a partition of the sample space $A_1, A_2, \ldots, A_k$,

$$P(A_i \mid B) = \frac{P(B \mid A_i)P(A_i)}{\sum_{j=1}^{k} P(B \mid A_j)P(A_j)}.$$

Example 9: Medical Test

A disease affects 1% of the population. A test is 95% accurate: if you have the disease, the test is positive with probability 0.95; if you don't, it is positive with probability 0.05. If a person tests positive, what is the probability they have the disease?
Let D = has disease, $+$ = positive test.

$$P(D) = 0.01, \quad P(D^c) = 0.99, \quad P(+ \mid D) = 0.95, \quad P(+ \mid D^c) = 0.05.$$

By Bayes:

$$P(D \mid +) = \frac{P(+ \mid D)P(D)}{P(+ \mid D)P(D) + P(+ \mid D^c)P(D^c)} = \frac{0.95 \times 0.01}{0.95 \times 0.01 + 0.05 \times 0.99} \approx 0.161.$$

Despite the test's accuracy, the probability is only about 16% because the disease is rare.

Tricks for Bayes Problems:

1. Identify the events: usually one is the "hypothesis" (what we want to know) and the other is the "evidence" (what we observe).
2. Write down the prior probabilities ($P(H)$) and the likelihoods ($P(E \mid H)$).
3. Use the law of total probability to compute $P(E)$.
4. Apply Bayes' formula.

Example 10: Two-Urn Problem

Urn 1 contains 2 red and 3 blue balls. Urn 2 contains 4 red and 1 blue ball. An urn is chosen at random, then a ball is drawn. Given that a red ball is drawn, what is the probability it came from Urn 1?
Let U_1 = choose Urn 1, U_2 = choose Urn 2, R = draw red.

$$P(U_1) = P(U_2) = 0.5, \quad P(R \mid U_1) = \frac{2}{5}, \quad P(R \mid U_2) = \frac{4}{5}.$$

$$P(U_1 \mid R) = \frac{P(R \mid U_1)P(U_1)}{P(R \mid U_1)P(U_1) + P(R \mid U_2)P(U_2)} = \frac{0.5 \times 2/5}{0.5 \times 2/5 + 0.5 \times 4/5} = \frac{1/5}{3/5} = \frac{1}{3}.$$

Summary of Key Formulas

Concept	Formula
Probability (equally likely)	$P(E) = \frac{\lvert E\rvert}{\lvert S\rvert}$
Multiplication Rule	$P(A \cap B) = P(A)P(B \mid A)$
Independence	$P(A \cap B) = P(A)P(B)$
Conditional Probability	$P(A \mid B) = \frac{P(A\cap B)}{P(B)}$
Law of Total Probability	$P(B) = \sum_i P(B \mid A_i)P(A_i)$
Bayes' Theorem	$P(A_i \mid B) = \frac{P(B\mid A_i)P(A_i)}{\sum_j P(B\mid A_j)P(A_j)}$
Permutations	$P(n,r) = \frac{n!}{(n-r)!}$
Combinations	$C(n,r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$

Table 1: Key Probability Formulas

These concepts form the basis for understanding uncertainty in data science. In machine learning, we often use conditional probability and Bayes' theorem in classifiers like Naïve Bayes, and counting techniques appear in evaluating model performance.